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LEARNER GUIDE

For

Fundamentals of AI Ethics and Responsible AI

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6.0	1 July 2026	Legacy courseware release.	Dr Alfred Ang
7.0	9 September 2026	Rebuilt as a visual, activity-led responsible AI course; added governance, data governance, GenAI security, copyright and meaningful human oversight.	Dr Alfred Ang

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How to Use This Guide

This guide contains the detailed procedures that are intentionally omitted from the course slides. Complete the activities in sequence using only the supplied mock data. Keep a copy of each deliverable as evidence for your learning portfolio.

Course Learning Outcomes

- LO1: Identify AI ethics and governance principles.
- LO2: Apply an AI ethics and governance framework to AI projects.

Skills Framework Coverage

- K1: AI Ethics and Governance frameworks
- K2: AI Ethics and Governance principles
- K3: AI Ethics and Governance processes
- K4: Relevant code of conduct for AI Ethics and Governance
- A1: Identify AI Ethics and Governance principles and processes
- A2: Uphold and comply with relevant AI Ethics and Governance pronouncements or code of conduct and ethics during performance of individual duties and responsibilities
- A3: Identify the linkages and apply the growing importance of AI Ethics and Governance to govern its deployment in AI-related projects

Course Proposal Constructive Alignment

Learning Unit	Learning Outcome	TSC statements	Modules	Assessment
LU1 · Introduction	LO1 · Identify principles	K1, K2, A1	Day 1 09:00–17:00 excl. lunch · 7h	WA Q1–Q2; CS Q1
LU2 · Applying frameworks	LO2 · Apply framework	K3, K4, A2, A3	Day 1 17:00–18:00 + Day 2 09:00–16:00 excl. lunch · 7h	WA Q3–Q4; CS Q2–Q3

Before You Start

- A PDF reader, spreadsheet application and word processor.
- Access to the LMS/TMS portal and the learner-safe GitHub repository.
- Use only the supplied mock data; never upload personal, confidential or client information to a public AI service.
- If an AI service is unavailable, perform the activity as a tabletop exercise using the prompts and templates.

Module 1 — AI Ethics Foundations (K1)

From AI capability to sociotechnical impact

What AI systems actually do

- **Perceive:** Extract patterns from images, sound, text and sensor data.
- **Predict:** Estimate a label, score, probability or future outcome.
- **Recommend:** Rank options or propose a course of action.
- **Generate:** Produce new text, images, audio, video or code probabilistically.

Key takeaway: Ethical risk follows the decision and impact, not the marketing label attached to the model.

Why AI ethics is operational

- **Scale:** A single defect can affect thousands of people before anyone notices.
- **Opacity:** Complex pipelines can hide why a result changed or who is accountable.
- **Power:** AI can shift opportunities, burdens and decision authority.
- **Feedback:** Predictions influence behaviour and create the next training data.

Key takeaway: AI ethics is how an organisation turns values into decisions, controls, evidence and redress.

The AI system is sociotechnical

- **People:** Owners, developers, users, reviewers and affected persons.
- **Process:** Policies, approvals, escalation routes and operating procedures.
- **Technology:** Data, models, prompts, tools, interfaces and infrastructure.
- **Context:** Law, culture, sector, incentives and the consequences of failure.

Key takeaway: A model can be technically accurate while the overall system remains harmful or unfair.

Risk, harm and benefit

- **Risk:** Likelihood multiplied by severity, adjusted for exposure and control strength.
- **Harm:** Impact on rights, safety, opportunity, dignity, privacy or the environment.
- **Benefit:** Value to individuals, organisations and society—not only efficiency.
- **Trade-off:** A justified balance with documented residual risk and accountable approval.

Key takeaway: Responsible AI seeks proportionate controls, not zero risk and not innovation at any cost.

Three ethical lenses

- **Consequences:** Which option produces the greatest overall benefit and least harm?
- **Duties and rights:** What obligations and non-negotiable rights must be respected?
- **Virtues and values:** What would a trustworthy, fair and responsible organisation do?
- **Public reason:** Could the decision and rationale be explained to affected people?

Key takeaway: Using more than one lens exposes blind spots that a single moral theory can miss.

Responsible AI as a control system

- **Intent:** Define purpose, intended users, prohibited uses and success criteria.
- **Constraints:** Set policy, legal, risk and human-oversight boundaries.

- **Evidence:** Test performance, fairness, robustness, privacy and security.
- **Feedback:** Monitor outcomes, incidents, drift, complaints and corrective action.

Key takeaway: A principle becomes real only when it has an owner, control, metric and evidence trail.

AI incidents have multiple causes

- **Data:** Unrepresentative samples, weak labels, leakage or unlawful collection.
- **Model:** Poor calibration, brittleness, opacity or unsafe generalisation.
- **Interface:** Automation bias, misleading confidence or no meaningful choice.
- **Governance:** No owner, weak monitoring, unclear escalation or ignored warnings.

Key takeaway: Investigate the whole chain; blaming the algorithm alone prevents durable remediation.

Stakeholder impact map

- **Decision makers:** Sponsor, product owner and accountable executive.
- **Operators:** Developers, vendors, reviewers and frontline staff.
- **Affected people:** Applicants, customers, employees and communities.
- **Assurance actors:** Risk, legal, security, audit, regulators and civil society.

Key takeaway: Those who bear the risk may not be the same people who receive the benefit.

Activity 1 — Map an AI Use Case and Its Stakeholders

Goal: Identify the system purpose, decision, stakeholders, benefits, harms and ethical tensions (A1).

Deliverable: Completed AI Use-Case & Stakeholder Canvas

Folder: labs/Activity-01-map-an-ai-use-case-and-its-stakeholders/

Step-by-step

1. Open scenario.md and underline the AI-assisted decision, not merely the model output.
2. Complete stakeholder-map.csv with decision makers, operators, affected people and assurance actors.
3. For each affected group, record benefit, harm, severity, reversibility and current control.
4. Use Prompt 1 to test completeness; verify every suggestion against the scenario and label assumptions.
5. Use Prompt 2 to identify a power imbalance and add a feedback or contest route.
6. Write three prohibited uses and three measurable success conditions in the canvas.

Prompts

Act as an AI governance analyst. From the scenario, list the AI-assisted decision, intended users, affected people, benefits, foreseeable harms, and information gaps. Do not invent facts; label assumptions.

Challenge the stakeholder map. Identify one stakeholder with high exposure but low decision power, and propose one safeguard and one feedback channel for that stakeholder.

Acceptance check

The canvas names the decision, at least six stakeholders, two benefits, four harms, three prohibited uses and an accountable owner.

- ☐ AI-assisted decision is explicit
- ☐ Affected people are separate from users
- ☐ Benefits and harms name who experiences them
- ☐ Assumptions are labelled
- ☐ Prohibited uses and owner are present

Module 2 — Responsible AI Principles (K2)

Translate values into observable behaviour

Principles form a connected system

- **Fairness:** Comparable people receive equitable treatment and outcomes.
- **Transparency:** People know when AI is used, for what purpose and by whom.
- **Accountability:** Named owners remain answerable for decisions and controls.
- **Human-centricity:** Human well-being, agency and safety remain primary.

Key takeaway: Principles reinforce each other: explainability supports contestability, which supports accountability.

Fairness is a design decision

- **Define groups:** Select protected or vulnerable groups relevant to the context.
- **Define outcome:** State what favourable and harmful outcomes mean.
- **Choose metric:** Pick a fairness measure that matches the decision pathway.
- **Accept trade-offs:** Document who benefits, who is burdened and why.

Key takeaway: There is no universal fairness metric; the defensible choice begins with context and harm.

Transparency has layers

- **System notice:** Disclose that AI is involved and what role it plays.
- **Process clarity:** Explain inputs, workflow, controls and decision authority.
- **Technical evidence:** Provide model cards, data sheets, tests and audit logs.
- **Outcome reason:** Give the affected person a useful reason and next step.

Key takeaway: Transparency is audience-specific: a user needs clarity; an auditor needs traceable evidence.

Explainability, interpretability, transparency

- **Interpretability:** How directly a human can understand the model mechanism.
- **Explainability:** A post-hoc account of factors influencing a result.
- **Transparency:** Information about purpose, data, process, limits and ownership.
- **Contestability:** A practical way to question, correct or appeal a decision.

Key takeaway: An explanation is only useful when it supports an action, not merely curiosity.

Privacy and data stewardship

- **Purpose:** Collect and use data for clear, legitimate and compatible purposes.
- **Minimisation:** Use only what is necessary for the stated objective.
- **Protection:** Apply access, encryption, retention and disclosure controls.
- **Agency:** Enable notice, consent where appropriate, correction and withdrawal.

Key takeaway: Privacy is a lifecycle obligation, not a one-time consent checkbox.

Safety, security and robustness

- **Safety:** Prevent unacceptable physical, psychological or societal harm.
- **Security:** Protect confidentiality, integrity and availability against threats.
- **Robustness:** Maintain acceptable behaviour under variation and attack.

- **Resilience:** Detect failure, contain impact and recover safely.

Key takeaway: Security protects the system; safety protects people; responsible AI requires both.

Accountability needs evidence

- **Owner:** A named role accepts responsibility for the system and outcome.
- **Decision record:** Approvals, assumptions, exceptions and residual risk are logged.
- **Assurance:** Independent review tests whether controls work as claimed.
- **Remedy:** Affected people have a route to correction, appeal and redress.

Key takeaway: If nobody can show the decision record, accountability exists only on paper.

Principles to policy to practice

- **Principle:** A durable value such as fairness or human agency.
- **Policy:** A mandatory organisational rule expressing that value.
- **Control:** A preventive, detective or corrective mechanism.
- **Evidence:** A test result, log, approval, record or metric proving operation.

Key takeaway: Write every code-of-conduct statement so that compliance can be observed and audited.

Activity 2 — Turn Principles into an AI Code of Conduct

Goal: Translate responsible AI principles into observable duties, controls and evidence (A1, A2).

Deliverable: AI Ethics Code of Conduct with control statements

Folder: labs/Activity-02-turn-principles-into-an-ai-code-of-conduct/

Step-by-step

1. Review principles.csv and choose the eight principles relevant to BrightHire.
2. Write one mandatory duty for each principle using clear 'must' language.
3. Add a preventive, detective or corrective control and name its control owner.
4. Define evidence such as an approval, test report, decision log, model card or complaint record.
5. Use the audit prompt to identify vague statements; revise any item that cannot be tested.
6. Record one exception process and one consequence for deliberate non-compliance.

Prompts

Rewrite each broad AI ethics principle as a mandatory policy statement using 'must'. Add an owner, trigger, control and evidence artefact. Keep it specific enough to audit.

Act as an internal auditor. For each policy statement, explain what evidence would prove compliance and what failure would look like.

Acceptance check

Every principle has a mandatory duty, owner, control, frequency/trigger, evidence and failure indicator.

- ☐ Statements use must, not should
- ☐ Each control has one owner
- ☐ Evidence is named
- ☐ Exception route is documented
- ☐ Human agency and redress are covered

Module 3 — AI Governance Frameworks (K1 • K2 • A1)

Use complementary frameworks without creating checklist theatre

Singapore Model AI Governance Framework

- **Internal governance:** Clear roles, responsibilities, SOPs, training and risk ownership.
- **Human involvement:** Set the appropriate degree of human control by harm and reversibility.
- **Operations:** Manage data, bias, explainability, robustness and model maintenance.
- **Stakeholder interaction:** Communicate clearly and provide feedback or contest routes.

Key takeaway: Singapore's framework translates explainable, transparent, fair and human-centric principles into practice.

AI Verify assurance model

- **Governance tests:** Document processes, oversight, accountability and transparency.
- **Technical tests:** Measure fairness, robustness, explainability and reproducibility.
- **Evidence:** Combine test outputs with context, limitations and human judgement.
- **Communication:** Produce a transparent report for internal and external stakeholders.

Key takeaway: Testing supports—not replaces—the organisation's responsibility to interpret results and manage risk.

NIST AI RMF core

- **Govern:** Create culture, roles, policy, risk tolerance and accountability.
- **Map:** Understand context, people, intended use, impacts and dependencies.
- **Measure:** Evaluate trustworthiness, uncertainty, performance and control effectiveness.
- **Manage:** Prioritise, treat, monitor and communicate AI risks.

Key takeaway: Govern is cross-cutting; Map, Measure and Manage iterate throughout the lifecycle.

ISO/IEC 42001 management system

- **Plan:** Establish context, policy, objectives, risks and AI impact processes.
- **Do:** Operate controls, competence, communication and documented processes.
- **Check:** Monitor, measure, audit and conduct management review.
- **Act:** Correct nonconformities and continually improve the AI management system.

Key takeaway: ISO 42001 governs the organisational system around AI, not the technical quality of one model alone.

Frameworks serve different questions

- **Singapore MGF:** How should an organisation operationalise trustworthy AI?
- **AI Verify:** How can governance and technical claims be tested and evidenced?
- **NIST AI RMF:** How do we structure continuous AI risk management?
- **ISO 42001:** How do we manage and continually improve an enterprise AIMS?

Key takeaway: Crosswalk overlapping requirements once; do not make teams complete four disconnected checklists.

AI governance operating model

- **Board and executive:** Set values, risk appetite, accountability and resources.
- **AI governance council:** Approve high-risk uses, exceptions and remediation.

- **Product and engineering:** Build controls, tests, documentation and monitoring.
- **Independent assurance:** Challenge evidence and verify control effectiveness.

Key takeaway: The first line owns the risk; the second guides and challenges; the third assures independently.

Use-case inventory and risk tiers

- **Inventory:** Record owner, purpose, users, data, model, vendor and deployment state.
- **Inherent risk:** Assess impact before controls using harm, scale and vulnerability.
- **Control profile:** Match testing, approvals and oversight to the risk tier.
- **Residual risk:** Approve, reduce, transfer, avoid or monitor after controls.

Key takeaway: No inventory means no reliable governance: unknown AI cannot be assessed, monitored or retired.

Governance anti-patterns

- **Policy-only:** A principle document exists but product teams lack controls and evidence.
- **Committee bottleneck:** Every decision waits for one central group regardless of risk.
- **Vendor blind trust:** Contract language substitutes for technical and operational assurance.
- **Launch-and-forget:** Approval occurs once with no monitoring, incident route or review trigger.

Key takeaway: Effective governance is risk-based, distributed, evidence-led and continuous.

Activity 3 — Tier AI Risk and Build a Governance RACI

Goal: Apply governance frameworks to classify AI use cases and assign decision rights (A1, A3).

Deliverable: AI inventory, risk-tier rationale and governance RACI

Folder: labs/Activity-03-tier-ai-risk-and-build-a-governance-raci/

Step-by-step

1. Open ai-inventory.csv and confirm each row has an owner, purpose, data, model/provider and deployment state.
2. Score severity, scale, reversibility, affected-person vulnerability and autonomy from 1 to 4.
3. Calculate inherent risk and assign low, medium, high or prohibited with a written rationale.
4. Map the Singapore MGF, NIST AI RMF and ISO 42001 practices to one shared control set.
5. Complete raci.csv for seven lifecycle decisions and check for one Accountable owner per row.
6. Define approval thresholds and review frequency for each tier.

Prompts

Using severity, scale, reversibility, vulnerability and decision autonomy, recommend a risk tier for each use case. Explain each score and identify missing information.

Create a RACI proposal for use-case approval, data approval, validation, deployment, monitoring, incident response and retirement. Ensure exactly one Accountable role per decision.

Acceptance check

All use cases have a defensible tier, minimum control profile, review cadence and unambiguous RACI.

- ☐ Inventory is complete
- ☐ Scores have reasons
- ☐ Prohibited-use rule exists
- ☐ One Accountable role per decision
- ☐ Framework controls are crosswalked

Module 4 — AI Governance Lifecycle (K3 · A1 · A3)

Control the system from idea to retirement

Lifecycle governance gates

- **Discover:** Clarify need, alternatives, stakeholders and prohibited uses.
- **Design:** Select data, model, controls, oversight and acceptance criteria.
- **Validate:** Test effectiveness, fairness, privacy, security and usability.
- **Operate:** Monitor, respond, change-control and retire with evidence.

Key takeaway: A gate is a decision with entry criteria, evidence, accountable approver and recorded outcome.

AI impact assessment

- **Context:** Purpose, decision, scale, affected people and deployment environment.
- **Impact:** Rights, safety, opportunity, privacy, dignity and environmental effects.
- **Controls:** Prevention, detection, response, human oversight and communication.
- **Decision:** Residual risk, conditions, approvals, review date and owner.

Key takeaway: The assessment should change the design; a form completed after launch is not risk management.

Risk statement anatomy

- **Cause:** Because of a data, model, user, vendor or process weakness...
- **Event:** ...the AI system may produce or enable an unsafe outcome...
- **Impact:** ...leading to a defined harm for specific stakeholders.
- **Control gap:** Existing controls may fail because their coverage or evidence is weak.

Key takeaway: Specific risk statements lead to testable controls; vague labels such as 'bias risk' do not.

Control design

- **Prevent:** Block prohibited data, unsafe actions or unapproved deployments.
- **Detect:** Monitor drift, anomalies, harmful outputs, complaints and access.
- **Respond:** Pause, contain, correct, notify, investigate and restore safely.
- **Recover:** Rollback, retrain, revalidate and confirm residual risk.

Key takeaway: Use defence in depth; a single filter or reviewer is a fragile point of failure.

Validation evidence stack

- **Technical:** Performance, fairness, robustness, privacy and security test results.
- **Human factors:** Comprehension, workload, automation bias and accessibility tests.
- **Operational:** Runbooks, logging, fallback, incident and business-continuity exercises.
- **Governance:** Approvals, exceptions, model/data cards, contracts and audit evidence.

Key takeaway: A benchmark score is necessary evidence, but never sufficient evidence for responsible deployment.

Production monitoring

- **Inputs:** Population shift, missingness, sensitive data and prompt patterns.
- **Outputs:** Error, toxicity, fairness gaps, confidence and unsafe actions.

- **Outcomes:** Real-world benefit, harm, complaints, appeals and override rates.
- **Controls:** Guardrail coverage, reviewer quality, alert latency and remediation time.

Key takeaway: Monitor the consequences people experience, not only model accuracy.

AI incident response

- **Triage:** Identify affected systems, users, severity and active harm.
- **Contain:** Pause features, revoke access, isolate data or route to humans.
- **Investigate:** Preserve prompts, versions, logs, approvals and decision context.
- **Learn:** Correct root causes, notify stakeholders and update controls.

Key takeaway: AI incidents demand both technical response and a human-impact response.

Change and retirement

- **Triggers:** New model, data, prompt, vendor, use, market or legal requirement.
- **Revalidation:** Re-run tests proportional to the change and its risk.
- **Rollback:** Keep a tested route to the last approved safe version.
- **Retirement:** Archive evidence, delete data appropriately and inform dependent users.

Key takeaway: A material change creates a new risk decision; version numbers alone do not preserve assurance.

Activity 4 — Complete an AI Impact Assessment and Risk Register

Goal: Apply the governance lifecycle to assess impact, select controls and make a recorded go/no-go decision (A1, A3).

Deliverable: AI impact assessment, risk register and stage-gate decision

Folder: labs/Activity-04-complete-an-ai-impact-assessment-and-risk-register/

Step-by-step

1. Read deployment-change.md and record purpose, scope, scale, users, affected people and alternatives.
2. Write at least six cause-event-impact statements in risk-register.csv.
3. Score likelihood and severity before controls and identify the risk owner.
4. Add preventive, detective, responsive and recovery controls with evidence and test frequency.
5. Re-score residual risk and identify conditions, exceptions and a review date.
6. Write a go, conditional-go or no-go decision signed by the accountable owner.

Prompts

Draft cause-event-impact risk statements for the scenario. Each statement must name the affected stakeholder and avoid vague labels such as 'bias risk'.

Challenge the control plan using defence in depth. Identify single points of failure and propose preventive, detective, responsive and recovery controls.

Acceptance check

The assessment connects context, harms, controls, evidence, residual risk and an accountable decision with conditions.

- ☐ Alternatives considered
- ☐ Six specific risk statements
- ☐ Controls cover four modes
- ☐ Residual risk is rescored
- ☐ Decision and review date recorded

Module 5 — Fairness and Explainability (K3 • K4 • A1)

Measure disparities and explain decisions without false certainty

Bias enters at every stage

- **Historical:** Past inequity is encoded in outcomes used as labels.
- **Representation:** Some groups or contexts are absent or under-sampled.
- **Measurement:** Proxies and labels differ in quality across groups.
- **Deployment:** Users apply a model outside its tested context or over-trust it.

Key takeaway: Removing a protected field does not remove proxies, structural inequity or deployment bias.

Read a confusion matrix

- **True positive:** A positive case is correctly identified.
- **False positive:** A negative case is incorrectly flagged positive.
- **True negative:** A negative case is correctly identified.
- **False negative:** A positive case is missed—often the costliest error.

Key takeaway: Fairness analysis begins by asking which error creates harm for whom.

Fairness metrics

- **Demographic parity:** Positive decision rates are comparable across groups.
- **Equal opportunity:** True-positive rates are comparable across groups.
- **Equalised odds:** Both true-positive and false-positive rates are comparable.
- **Calibration:** A given score has comparable meaning across groups.

Key takeaway: Metrics can conflict; select and justify the one aligned to the use case and harm model.

Fairness dashboard

- **Selection rate:** Group A 68% versus Group B 52% indicates outcome disparity.
- **True-positive rate:** Group A 82% versus Group B 70% indicates missed opportunity.
- **False-positive rate:** Group A 12% versus Group B 22% indicates excess burden.
- **Response:** Investigate data, thresholds, features and process—not just the metric.

Key takeaway: Indicative teaching data: a disparity is a signal for investigation, not automatic proof of discrimination.

White-box and black-box explanations

- **White-box:** Rules, linear models and small trees can be interpreted directly.
- **Black-box:** Complex ensembles and deep models usually need post-hoc methods.
- **Global:** Describe overall behaviour, influential features and population patterns.
- **Local:** Explain why one prediction changed for one instance.

Key takeaway: Model performance and interpretability are not a binary choice; choose evidence proportional to risk.

LIME, SHAP and PDP

- **LIME:** Fits a simple local surrogate around one prediction.
- **SHAP:** Allocates contribution using Shapley-value reasoning.

- **PDP:** Shows the average marginal relationship between a feature and prediction.
- **Caution:** Correlation, unstable explanations and feature dependence can mislead.

Key takeaway: Explainability methods reveal model behaviour; they do not prove causality or fairness.

Counterfactual explanations

- **Question:** What minimal change would alter this outcome?
- **Use:** Give an affected person a concrete, actionable reason.
- **Constraint:** Changes must be feasible, lawful and not shift burden unfairly.
- **Risk:** A technically valid counterfactual may recommend an impossible life change.

Key takeaway: A useful explanation respects human agency and the real-world ability to act.

Documentation for explainability

- **Model card:** Intended use, performance, limits, groups and evaluation results.
- **Data sheet:** Provenance, consent, composition, processing and known gaps.
- **System card:** End-to-end behaviour, safeguards, tests and deployment context.
- **Decision notice:** Plain-language reason, human contact and contest route.

Key takeaway: Match documentation to audience while keeping every claim traceable to evidence.

Activity 5 — Audit Fairness and Explainability

Goal: Compute group-level fairness indicators, interpret explanations and recommend proportionate mitigations (A1, A2).

Deliverable: Fairness scorecard and explanation review

Folder: labs/Activity-05-audit-fairness-and-explainability/

Step-by-step

1. Open fairness-data.csv and create a confusion matrix for groups A and B.
2. Compute selection rate, true-positive rate, false-positive rate and precision for each group.
3. Calculate absolute gaps and describe which error creates the most serious harm.
4. Use the fairness prompt to propose data, threshold, feature and process investigations.
5. Review explanation-samples.csv and rate each explanation for fidelity, clarity and actionability.
6. Write a recommendation with chosen metric, limitations, mitigation, owner and monitoring threshold.

Prompts

Review the group metrics as an AI fairness analyst. Distinguish observation from conclusion, identify the harmed pathway, and propose investigations before recommending a fix.

Evaluate each explanation for fidelity, clarity, actionability and risk of misleading the affected person. Suggest a safer replacement notice.

Acceptance check

The scorecard shows correct metrics by group, interprets harm cautiously and connects mitigation to an owner and monitoring rule.

- ☐ Confusion matrices reconcile
- ☐ At least four metrics per group
- ☐ Metric choice is justified
- ☐ No causal claim from correlation
- ☐ Explanation supports contestability

Module 6 — Data Governance and Privacy (K4 • A2)

Make provenance, quality, access and retention governable

Data governance for AI

- **Ownership:** Data owner approves purpose, access, quality thresholds and retention.
- **Provenance:** Record source, collection method, licence, consent and transformations.
- **Quality:** Measure completeness, accuracy, timeliness, consistency and representativeness.
- **Control:** Enforce classification, access, sharing, retention and deletion.

Key takeaway: A model is only as governable as the evidence chain behind its data.

AI data lifecycle

- **Collect:** Confirm purpose, authority, minimisation and notice.
- **Prepare:** Profile, clean, label, document and protect sensitive fields.
- **Use:** Restrict features and access to the approved purpose.
- **Retain or delete:** Apply schedule, legal hold, lineage and verified disposal.

Key takeaway: Each hand-off needs a recorded owner, allowed purpose and control state.

Data lineage

- **Source:** Where did the data originate and under what authority?
- **Transform:** Which joins, filters, labels and feature engineering changed it?
- **Model:** Which dataset version trained or evaluated which model version?
- **Outcome:** Which data and model version influenced a specific decision?

Key takeaway: Lineage turns an unrepeatable result into an auditable decision chain.

Data quality is contextual

- **Accuracy:** Values reflect the real-world attribute closely enough for purpose.
- **Completeness:** Missingness is measured overall and by relevant group.
- **Representativeness:** Training data covers the population and conditions of use.
- **Freshness:** Update frequency matches how quickly the phenomenon changes.

Key takeaway: A dataset can be clean yet unfit for the decision it is being asked to support.

Privacy threat patterns

- **Linkage:** Quasi-identifiers reconnect de-identified records to individuals.
- **Membership inference:** An attacker infers whether a person was in training data.
- **Model inversion:** Outputs reveal sensitive attributes or representative records.
- **Prompt leakage:** Users expose personal or confidential data to a generative system.

Key takeaway: De-identification reduces risk; it does not automatically make data anonymous or unrestricted.

Privacy-enhancing techniques

- **K-anonymity:** Generalise quasi-identifiers so each record resembles at least $k-1$ others.
- **Differential privacy:** Bound an individual's influence using calibrated randomness.
- **Federated learning:** Train across locations without centralising raw data.
- **Synthetic data:** Generate representative records while testing disclosure and bias risk.

Key takeaway: Choose a technique by threat model; every privacy method introduces utility and implementation trade-offs.

Access, retention and deletion

- **Least privilege:** Grant only the data and action needed for the role.
- **Separation:** Keep production data away from experimentation and public tools.
- **Retention:** Set period, trigger, owner and approved exceptions.
- **Deletion:** Remove copies, embeddings, caches and backups where required.

Key takeaway: A deleted source file may survive in a vector store, fine-tune, cache or derived feature table.

Data governance evidence

- **Data inventory:** Dataset, owner, purpose, sensitivity, location and retention.
- **Data sheet:** Composition, collection, preprocessing, limitations and licences.
- **Quality report:** Thresholds, exceptions and group-level results.
- **Access log:** Who accessed which data, why, when and what changed.

Key takeaway: Evidence should allow an independent reviewer to reconstruct the data decision.

Activity 6 — Design Data Governance and Privacy Controls

Goal: Review AI data lineage, quality, privacy threats, access and retention controls (A2).

Deliverable: Data sheet, lineage map and privacy control plan

Folder: labs/Activity-06-design-data-governance-and-privacy-controls/

Step-by-step

1. Open data-inventory.csv and confirm owner, source, authority, sensitivity, purpose, location and retention.
2. Complete lineage.csv from source through transformations, training, evaluation and production decisions.
3. Measure missingness and representativeness overall and by group; record exceptions.
4. Threat-model linkage, membership inference, model inversion and prompt disclosure.
5. Select minimisation, access, encryption, k-anonymity, differential privacy or federated controls only where justified.
6. Define retention and deletion for source data, features, embeddings, logs, caches and backups.

Prompts

From the data inventory, identify purpose-limitation, consent, minimisation, quality, linkage, retention and access risks. Separate known facts from questions for the data owner.

Recommend privacy-enhancing techniques only where they match the threat model. State the utility cost, residual risk and evidence needed to validate each technique.

Acceptance check

The data plan can reconstruct provenance and sets measurable quality, access, privacy, retention and deletion requirements.

- ☐ Purpose and authority recorded
- ☐ Lineage reaches model decision
- ☐ Quality measured by group
- ☐ PET maps to threat
- ☐ Derived copies included in deletion

Module 7 — Generative AI Security and Copyright (K4 · A2)

Secure the data-model-application-human stack

Generative AI expands the attack surface

- **Data plane:** Training, fine-tuning, RAG and conversation data can be poisoned or leaked.
- **Model plane:** Weights, prompts, adapters and model endpoints can be stolen or manipulated.
- **Application plane:** Tools, plugins, parsers and output consumers inherit unsafe content.
- **Human plane:** Users may over-trust fluent output or disclose sensitive information.

Key takeaway: Treat the generative AI service as an untrusted component inside a controlled system.

OWASP LLM Top 10: input and data

- **Prompt injection:** Direct or indirect instructions override intended behaviour.
- **Sensitive disclosure:** The system reveals private, confidential or regulated information.
- **Supply chain:** Models, datasets, packages and providers introduce hidden risk.
- **Data/model poisoning:** Manipulated data or artefacts corrupt behaviour or create backdoors.

Key takeaway: Input filtering alone cannot solve prompt injection; constrain privileges, data and downstream actions.

OWASP LLM Top 10: output and agency

- **Improper output handling:** Unsafe model output reaches code, queries, browsers or users.
- **Excessive agency:** The model can perform actions beyond minimum necessary privileges.
- **System prompt leakage:** Hidden instructions are exposed and mistaken for a security boundary.
- **Unbounded consumption:** Inputs or loops exhaust cost, compute, time or service capacity.

Key takeaway: Validate model output at the trust boundary and require approval for high-impact actions.

RAG and vector-store risks

- **Poisoned documents:** Malicious content steers retrieval or injects hidden instructions.
- **Access bleed:** Chunks from one user, tenant or classification reach another.
- **Embedding leakage:** Vector representations can expose sensitive relationships.
- **Stale knowledge:** Outdated or revoked documents remain retrievable.

Key takeaway: Carry identity, permissions, provenance and expiry from source document to every chunk and retrieval.

Layered generative AI controls

- **Before the model:** Classify input, remove secrets, enforce policy and retrieve authorised data.
- **At the model:** Use bounded instructions, safer settings and model/provider assurance.
- **After the model:** Validate claims, content, format and tool arguments before use.
- **Around the system:** Least privilege, approvals, logging, monitoring, rate limits and response.

Key takeaway: Defence in depth assumes every individual guardrail will eventually fail.

Hallucination and misinformation

- **Cause:** The model generates probable language, not verified truth.

- **Exposure:** Open-ended questions and missing evidence increase uncertainty.
- **Control:** Ground responses, cite sources, verify claims and calibrate refusal.
- **Oversight:** Route consequential content to qualified human review.

Key takeaway: Fluency is not evidence; require provenance for claims that drive decisions.

Copyright across the GenAI lifecycle

- **Training input:** Check source rights, licences, consent, exceptions and contractual limits.
- **Prompt input:** Do not upload protected or confidential content without authority.
- **Generated output:** Assess similarity, attribution, ownership terms and intended use.
- **Publication:** Keep provenance records and review high-risk commercial outputs.

Key takeaway: Copyright risk depends on jurisdiction and facts; governance should preserve provenance and enable review.

Security architecture evidence

- **Threat model:** Assets, actors, trust boundaries, abuse cases and mitigations.
- **Evaluation:** Prompt-attack, leakage, output, access and cost tests.
- **Telemetry:** Prompt category, retrieval source, tool call, approval and outcome logs.
- **Runbook:** Containment, key rotation, corpus purge, rollback and notification.

Key takeaway: A secure design can show which layer stops each threat and what evidence proves it.

Activity 7 — Threat-Model a Generative AI Assistant

Goal: Apply layered generative AI security, privacy and copyright controls (A2).

Deliverable: Threat model, adversarial prompt results and control architecture

Folder: labs/Activity-07-threat-model-a-generative-ai-assistant/

Step-by-step

1. Open architecture.json and draw the users, application, model, vector store, tools and logging trust boundaries.
2. Map applicable OWASP 2025 risks and the assets each risk threatens.
3. Use only the benign attack prompts in adversarial-prompts.txt against a sandbox or tabletop scenario.
4. Record expected safe behaviour, observed behaviour and evidence without entering real secrets.
5. Design controls before, at, after and around the model; apply least privilege to every tool.
6. Add copyright/provenance checks and write a contain-investigate-recover incident runbook.

Prompts

Act as a defensive GenAI security reviewer. Map the scenario to the OWASP Top 10 for LLM Applications 2025. For each applicable risk, name the trust boundary, abuse case, impact and defence-in-depth controls.

Review the proposed control architecture for prompt injection, sensitive disclosure, excessive agency, improper output handling, poisoned retrieval and unbounded consumption. Identify any control that relies only on the model following instructions.

Acceptance check

The threat model covers data, model, application and human layers; every high-risk abuse case has multiple controls and evidence.

- ☐ Trust boundaries drawn
- ☐ OWASP risks mapped
- ☐ No real secrets used
- ☐ Tool permissions are least privilege
- ☐ Copyright and incident controls included

Module 8 — Human Oversight and Accountability (A2 · A3)

Keep meaningful human control where consequences matter

Human involvement models

- **Human-in-the-loop:** A person approves each consequential recommendation before action.
- **Human-on-the-loop:** AI acts within bounds while a person supervises and can intervene.
- **Human-out-of-the-loop:** AI acts without real-time review; reserve for low-risk, reversible tasks.
- **Human-in-command:** People set goals, limits, authority, escalation and shutdown conditions.

Key takeaway: Meaningful oversight depends on power, information, time and competence—not the presence of a checkbox.

Choose oversight by risk

- **Severity:** Greater potential harm requires stronger review and authority.
- **Reversibility:** Irreversible outcomes require intervention before action.
- **Uncertainty:** Novel contexts and weak confidence increase escalation.
- **Scale:** High-volume decisions need sampling, monitoring and systemic safeguards.

Key takeaway: Automate low-risk routine decisions; preserve human approval for high-impact exceptions.

Design a meaningful review

- **Information:** Show evidence, uncertainty, alternatives and policy—not only a score.
- **Time:** Give the reviewer enough time to understand and challenge the result.
- **Authority:** Allow override, pause, escalation and correction without penalty.
- **Competence:** Train reviewers and monitor the quality of their decisions.

Key takeaway: A reviewer who cannot understand or change the decision is not an effective control.

Automation bias and rubber-stamping

- **Anchoring:** The AI recommendation becomes the default starting point.
- **Complacency:** Repeated correct outputs reduce vigilance over time.
- **Asymmetry:** Overriding the model takes more effort or creates perceived blame.
- **Countermeasure:** Hide score initially, require reasons, sample reviews and measure overrides.

Key takeaway: Human review can create false assurance unless the interface and incentives support genuine challenge.

Decision rights and escalation

- **Operate:** Frontline users handle normal cases within approved thresholds.
- **Review:** Subject-matter experts resolve low-confidence and vulnerable-person cases.
- **Approve:** Accountable owners accept high residual risk and exceptions.
- **Stop:** Designated roles can pause the system when safety or rights are threatened.

Key takeaway: Every boundary needs a trigger, response time, owner and fallback.

RACI for AI governance

- **Responsible:** Performs the task and produces the evidence.
- **Accountable:** Owns the outcome and has final decision authority.

- **Consulted:** Provides specialist input before the decision.
- **Informed:** Receives the outcome and relevant changes.

Key takeaway: There should be one accountable owner for each material risk decision.

Audit trail and contestability

- **Trace:** Record input, model/data version, output, reviewer and final action.
- **Explain:** Provide a reason suited to the affected person's needs.
- **Correct:** Enable factual correction and rerun where appropriate.
- **Appeal:** Offer timely independent review and communicate the outcome.

Key takeaway: Redress completes accountability by giving affected people a path from explanation to remedy.

Governance maturity roadmap

- **Ad hoc:** Teams rely on individual judgement and scattered documents.
- **Defined:** Inventory, policy, roles, risk tiers and minimum controls exist.
- **Measured:** Control effectiveness, outcomes, incidents and fairness are monitored.
- **Adaptive:** Evidence drives continuous improvement, challenge and transparent reporting.

Key takeaway: Maturity is demonstrated by repeatable evidence and better outcomes—not by the number of policies.

Activity 8 — Design Human Oversight and a Governance Decision

Goal: Select meaningful human involvement, escalation, monitoring and redress for an AI deployment (A2, A3).

Deliverable: Oversight matrix and governance board decision pack

Folder: labs/Activity-08-design-human-oversight-and-a-governance-decision/

Step-by-step

1. Open decision-scenarios.csv and score severity, reversibility, uncertainty, scale and vulnerability.
2. Choose an oversight model and explain why weaker or stronger options are unsuitable.
3. Define information shown to the reviewer, authority to override/pause and maximum response time.
4. Add escalation triggers for low confidence, vulnerable people, complaints, drift and control failure.
5. Design notice, factual correction, independent appeal and remedy pathways.
6. Prepare and sign a board decision with conditions, monitoring thresholds, incident triggers and review date.

Prompts

For each decision scenario, recommend human-in-the-loop, human-on-the-loop or human-out-of-the-loop. Use severity, reversibility, uncertainty, scale and vulnerable-person impact. Define reviewer authority and response time.

Act as an AI governance council. Challenge the deployment proposal and write a go, conditional-go or no-go decision with evidence gaps, approval conditions, monitoring thresholds, incident triggers and a review date.

Acceptance check

The oversight design is risk-proportionate, gives reviewers real authority, addresses automation bias and includes contestability and stop conditions.

- ☐ Oversight choice is justified
- ☐ Reviewer has time and authority
- ☐ Escalation thresholds are measurable
- ☐ Appeal and correction exist
- ☐ Board decision records residual risk

Assessment Flow

1. TRAQOM — complete the required digital attendance.
2. Assessment Digital Attendance — submit the assessment attendance record.
3. Assessment — complete the 1-hour WA followed by the 1-hour Case Study.
4. Submit the completed question papers on the LMS.
5. Sign the Assessment Summary Record after feedback.

Reference Frameworks

- Singapore Model AI Governance Framework, 2nd ed.: <https://www.pdpc.gov.sg/help-and-resources/2020/01/model-ai-governance-framework>
- Singapore Model AI Governance Framework for Generative AI: <https://www.imda.gov.sg/resources/press-releases-factsheets-and-speeches/factsheets/2024/gen-ai-and-digital-foss-ai-governance-playbook>
- NIST AI Risk Management Framework 1.0: <https://www.nist.gov/itl/ai-risk-management-framework>
- NIST AI 600-1 Generative AI Profile: <https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf>
- OWASP Top 10 for LLM Applications 2025: <https://genai.owasp.org/llm-top-10/>
- ISO/IEC 42001:2023 overview: <https://www.iso.org/standard/42001>

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